



Gisma University
of Applied Sciences

Assessment Submission Form

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Assessment Title	Final Assessment
Module Code	B105
Module Title	Applied Statistical Modelling
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Date Submitted	25/09/2024

Declaration of Authorship

I declare that all material in this assessment is my own work except where there is clear acknowledgement and appropriate reference to the work of others.

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Signed.....*lana*..... Date25/09/2024.....

Lung Cancer Patient Analysis

Source: <https://www.kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link>

Github: <https://github.com/iLearex/first/blob/main/Rhistorystats.Rhistory>

<https://github.com/iLearex/first/blob/main/.Rhistory>

Introduction:

With the increasing popularity of cigarettes, e-cigarettes and/or vapes especially among young people, and with an increase in the lung cancer rates in women over the past 2-5 years, it is important to understand what factors contribute and affect lung cancer rates and the severity of the disease in diagnosed individuals (McGinnis, A. (2022).

Key Business Questions:

What is the relationship between the variables and the target label?

What factors affect the severity of cancer (Level) the most?

Are there specific factors that correlate strongly with each other?

Can we predict a patient's cancer severity or level based on the available factors?

Description:

The dataset that I have selected for my project is a table containing some demographic and medical information (26 columns) about 1,000 cancer patients (the rows). I will also refer to the patient information as factors.

The independent variables are:

- Index: An identifier (sequential integer).
- Patient Id: A unique identifier representing each patient.
- Age: Patient's age (integer).
- Gender: Coded as 1 (male), 0 (female).
- Air Pollution, Alcohol use, Dust Allergy, Occupational Hazards, Genetic Risk, Chronic Lung Disease, Balanced Diet, Obesity, Smoking, Passive Smoker, Chest Pain, Coughing of Blood, Fatigue, Weight Loss, Shortness of Breath, Wheezing, Swallowing Difficulty, Clubbing of Finger Nails, Frequent Cold, Dry Cough, Snoring: These are all integer values ranging from 1 to 10, representing different categorical levels of exposure or severity.

The target label:

- Level: indicating the severity of the cancer condition, categorized as Low, Medium, or High (categorical).

Data Preprocessing, sampling and cleaning:

(some code screenshots may be from R history, either on R Studio or VS but it's mostly from my R Studio project and console code. Both will be found on my GitHub)

The dataset is relatively clean, not too large and as will be clearly demonstrated later, is relatively balanced as well.

Step1:

Read, and view (named the dataset as df (for ease))

```
df <- read.csv("cancer patient data sets.csv")
```

Then I used view(df) to be able to see the dataset in R. After that, I ran str(df) to get an overview of the dataset, get the column names, and identify the different datatypes of each column. As shown in the code snippet below, all the columns have the same datatype (integers), except for Level which has the character datatype.

```
> str(df)
'data.frame': 1000 obs. of 24 variables:
 $ Age          : int  33 17 35 37 46 35 52 28 35 46 ...
 $ Gender       : int  1 1 1 1 1 1 2 2 2 1 ...
 $ Air.Pollution : int  2 3 4 7 6 4 2 3 4 2 ...
 $ Alcohol.use  : int  4 1 5 7 8 5 4 1 5 3 ...
 $ Dust.Allergy : int  5 5 6 7 7 6 5 4 6 4 ...
 $ Occupational.Hazards : int  4 3 5 7 7 5 4 3 5 2 ...
 $ Genetic.Risk : int  3 4 5 6 7 5 3 2 6 4 ...
 $ chronic.Lung.Disease : int  2 2 4 7 6 4 2 3 5 3 ...
 $ Balanced.Diet : int  2 2 6 7 7 6 2 4 5 3 ...
 $ Obesity      : int  4 2 7 7 7 7 4 3 5 3 ...
 $ Smoking      : int  3 2 2 7 8 2 3 1 6 2 ...
 $ Passive.Smoker : int  2 4 3 7 7 3 2 4 6 3 ...
 $ Chest.Pain    : int  2 2 4 7 7 4 2 3 6 4 ...
 $ Coughing.of.Blood : int  4 3 8 8 9 8 4 1 5 4 ...
 $ Fatigue       : int  3 1 8 4 3 8 3 3 1 1 ...
 $ Weight.Loss   : int  4 3 7 2 2 7 4 2 4 2 ...
 $ Shortness.of.Breath : int  2 7 9 3 4 9 2 2 3 4 ...
 $ Wheezing      : int  2 8 2 1 1 2 2 4 2 6 ...
 $ Swallowing.Difficulty : int  3 6 1 4 4 1 3 2 4 5 ...
 $ Clubbing.of.Finger.Nails : int  1 2 4 5 2 4 1 2 6 4 ...
 $ Frequent.Cold : int  2 1 6 6 4 6 2 3 2 2 ...
 $ Dry.Cough     : int  3 7 7 7 2 7 3 4 4 1 ...
 $ Snoring       : int  4 2 2 5 3 2 4 3 1 5 ...
 $ Level         : chr  "Low" "Medium" "High" "High" ...
```

Since I viewed this dataset after removing the columns Patient.Id and index, they are not shown in the above screenshot.

Step 2:

Remove unnecessary columns; in this case Patient.Id and index are not needed for the analysis, and do not contribute much, so I dropped them.

There are many libraries, for example dplyr, that can be used for dropping columns, but I just used the built-in R commands.

```
> view(df)
> df <- df[, !names(df) %in% c("Patient.Id")]
df <- df[, !names(df) %in% c("index")]
```

Or I also used subset and -c to select multiple columns and to delete both at the same time.

```
> df <- subset(df, select = -c(index, Patient.Id))
```

Step 3:

Checking for nulls

```
> sum(is.na(df))
[1] 0
```

I used the sum() function to get the total number of nulls within the dataset and identified that there are no nulls.

Step 4:

Encoding the categorical column 'Level'

As previously pointed out, Level had a character datatype but since we need all the columns to have numerical datatypes for the analysis, so I encoded it with the as.numeric function which automatically changed the words Low, Medium, and High into numerical values, but after I ran this function and saw the results, I realized that it is not ideal since it changed the words describing the Levels based on alphabetical order which was confusing.

```
$ Level      : chr "Low" "Medium" "High" "High" ...
> df$Level <- as.numeric(as.factor(df$Level))
```

Therefore; I opted for reloading the data frame and changing the Level words manually in a way that makes more sense.

```
> df <- read.csv("cancer patient data sets.csv")
> library(dplyr)
> df$Level <- recode(df$Level, "Low" = 1, "Medium" = 2, "High" = 3)
```

Descriptive Analysis:

Mean, Median and Standard deviation

First, I ran summary(df) which shows the minimum & maximum values, mean, median, and the 1st & 3rd quartile values. When looking at the returned values, we can see that the range of values for all the factors is 1-10, with none reaching ten. Of course, excluding Age since it has continuous integer values

and gender which is binary. I examined the mean and median values for each variable and they all are relatively balanced. It was interesting to see though, that the Age variable has some quite extreme values in the Min. And Max,

```
> summary(df)
      Age      Gender  Air.Pollution  Alcohol.use  Dust.Allergy
Min.   :14.00  Min.   :1.000  Min.   :1.00  Min.   :1.000  Min.   :1.000
1st Qu.:27.75  1st Qu.:1.000  1st Qu.:2.00  1st Qu.:2.000  1st Qu.:4.000
Median :36.00  Median :1.000  Median :3.00  Median :5.000  Median :6.000
Mean   :37.17  Mean   :1.402  Mean   :3.84  Mean   :4.563  Mean   :5.165
3rd Qu.:45.00  3rd Qu.:2.000  3rd Qu.:6.00  3rd Qu.:7.000  3rd Qu.:7.000
Max.   :73.00  Max.   :2.000  Max.   :8.00  Max.   :8.000  Max.   :8.000
Occupational.Hazards  Genetic.Risk  chronic.Lung.Disease  Balanced.Diet
Min.   :1.00  Min.   :1.00  Min.   :1.00  Min.   :1.000
1st Qu.:3.00  1st Qu.:2.00  1st Qu.:3.00  1st Qu.:2.000
Median :5.00  Median :5.00  Median :4.00  Median :4.000
Mean   :4.84  Mean   :4.58  Mean   :4.38  Mean   :4.491
3rd Qu.:7.00  3rd Qu.:7.00  3rd Qu.:6.00  3rd Qu.:7.000
Max.   :8.00  Max.   :7.00  Max.   :7.00  Max.   :7.000
Obesity      Smoking      Passive.Smoker      Chest.Pain      Coughing.of.Blood
Min.   :1.000  Min.   :1.000  Min.   :1.000  Min.   :1.000  Min.   :1.000
1st Qu.:3.000  1st Qu.:2.000  1st Qu.:2.000  1st Qu.:2.000  1st Qu.:3.000
Median :4.000  Median :3.000  Median :4.000  Median :4.000  Median :4.000
Mean   :4.465  Mean   :3.948  Mean   :4.195  Mean   :4.438  Mean   :4.859
3rd Qu.:7.000  3rd Qu.:7.000  3rd Qu.:7.000  3rd Qu.:7.000  3rd Qu.:7.000
Max.   :7.000  Max.   :8.000  Max.   :8.000  Max.   :9.000  Max.   :9.000
Fatigue      Weight.Loss      Shortness.of.Breath      Wheezing
Min.   :1.000  Min.   :1.000  Min.   :1.00  Min.   :1.000
1st Qu.:2.000  1st Qu.:2.000  1st Qu.:2.00  1st Qu.:2.000
Median :3.000  Median :3.000  Median :4.00  Median :4.000
Mean   :3.856  Mean   :3.855  Mean   :4.24  Mean   :3.777
3rd Qu.:5.000  3rd Qu.:6.000  3rd Qu.:6.00  3rd Qu.:5.000
Max.   :9.000  Max.   :8.000  Max.   :9.00  Max.   :8.000
Swallowing.Difficulty  Clubbing.of.Finger.Nails  Frequent.Cold  Dry.Cough
Min.   :1.000  Min.   :1.000  Min.   :1.000  Min.   :1.000
1st Qu.:2.000  1st Qu.:2.000  1st Qu.:2.000  1st Qu.:2.000
Median :4.000  Median :4.000  Median :3.000  Median :4.000
Mean   :3.746  Mean   :3.923  Mean   :3.536  Mean   :3.853
3rd Qu.:5.000  3rd Qu.:5.000  3rd Qu.:5.000  3rd Qu.:6.000
Max.   :8.000  Max.   :9.000  Max.   :7.000  Max.   :7.000
Snoring      Level
Min.   :1.000  Min.   :1.000
1st Qu.:2.000  1st Qu.:1.000
Median :3.000  Median :2.000
Mean   :2.926  Mean   :2.062
3rd Qu.:4.000  3rd Qu.:3.000
Max.   :7.000  Max.   :3.000
```

So I tested for outliers. I installed The Outliers package and used the Grubbs Test to identify the presence of any outliers. The Grubbs test is also useful because it provides other insights, such as p-value and suggests an alternate hypothesis for the outlier test. The results show that there is a single outlier for age and that is the age of 73.

```

> install.packages("outliers")
WARNING: Rtools is required to build R packages but is not currently installed. Please download
install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/lamat/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.4/outliers_0.15.zip'
Content type 'application/zip' length 84619 bytes (82 KB)
downloaded 82 KB

package 'outliers' successfully unpacked and MD5 sums checked

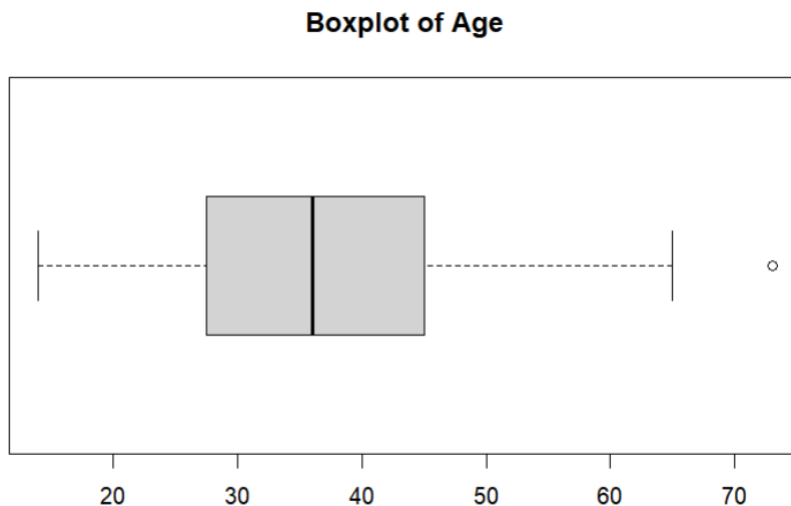
The downloaded binary packages are in
C:\Users\lamat\AppData\Local\Temp\RtmpIhMAXP\downloaded_packages
> library(outliers)
> grubbs.test(df$Age)

Grubbs test for one outlier

data: df$Age
G = 2.98413, U = 0.99108, p-value = 1
alternative hypothesis: highest value 73 is an outlier

```

I further demonstrated this by creating a boxplot. This plot also helps us visualize for the majority of the age values are.



The boxplot also shows the outlier as the circle on the far left, showing that it is a lot further than the range of values. We can also see that it is only a single value and the rest are relatively normally distributed thus, we can still consider the age as balanced since a single outlier would not have a huge effect on the results.

Is the target label balanced?

Since my target label is Level and it is important that it is balanced in order to have efficient, unbiased results I tested if it is balanced.

```
> table(df$Level)
```

```
 1    2    3
303 332 365
```

The data is relatively balanced as there is no huge difference in values between the different levels.

Hypothesis

I formulated multiple Hypotheses that I tested by using different Inferential statistical methods and tests.

H1: Higher levels of Air Pollution mean a more advanced Level of Cancer.

H2: Individuals exposed to more Occupational Hazards have higher Levels of Cancer.

H3: Patients that drink more have more severe Cancer levels.

H0: None of these factors are significantly positively correlated and do not seriously affect the Level of cancer.

Inferential Statistics

Correlation Analysis:

I tested to see the correlation between different variables and my target label. Testing for correlation will help us better understand which factors influence the Level/stage of cancer the most. First, I tested the correlation between Air pollution and Level. To do that, I installed some correlation packages, and the ggplot2 package to create a visualization that makes it easier to understand the relationship between the variable (Air.Pollution) and the target label (Level 'of cancer').

```
> library(corrplot)
> corrplot(cor_matrix, method = "circle")
> install.packages("ggplot2")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and i
n stall the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/lamat/AppData/Local/R/win-library/4.4'
(as 'lib' is unspecified)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.4/ggplot2_3.5.1.zip'
Content type 'application/zip' length 5012863 bytes (4.8 MB)
downloaded 4.8 MB

package 'ggplot2' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
  C:/Users/lamat/AppData/Local/Temp/RtmpIhMAXP/downloaded_packages
> library(ggplot2)
Use suppressPackageStartupMessages() to eliminate package startup messages

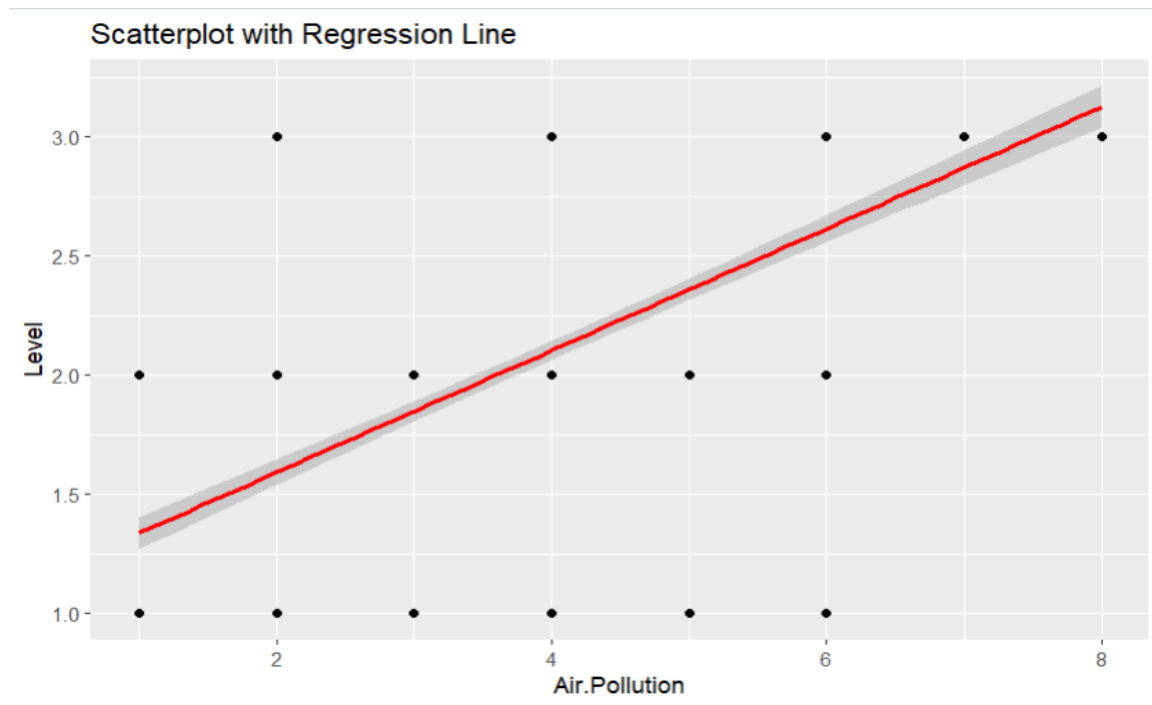
Attaching package: 'ggplot2'

The following objects are masked from 'package:psych':

  %+%, alpha

> ggplot(df, aes(x = Air.Pollution, y = Level)) +
+   geom_point() +
+   geom_smooth(method = "lm", col = "red") +
+   labs(title = "Scatterplot with Regression Line")
`geom_smooth()` using formula = 'y ~ x'
> |
```

GGplot creates a scatterplot with a regression line demonstrating the relationship between variable and target variable. As shown below and since the regression line (red line) fits well in the graph, there is a positive relatively strong linear relationship between Air.Pollution and The Level, which means that patients exposed to more air pollution would probably have more advanced levels of Lung cancer.



I used different correlation tests, such as 'Pearson' and I calculated the p-value to understand statistical significance of the correlation between Air pollution and cancer level. The p-value came out to be $< 2.2 \times 10^{-16}$, which is extremely small. That means the test is highly statistically significant.

```
> cor(df$Air.Pollution, df$Level, use = "complete.obs")
[1] 0.6360385
> cor(df$Air.Pollution, df$Level, method = "pearson")
[1] 0.6360385
> cor.test(df$Air.Pollution, df$Level, method = "pearson")
```

Pearson's product-moment correlation

```
data: df$Air.Pollution and df$Level
t = 26.039, df = 998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.5976091 0.6715523
sample estimates:
      cor
0.6360385
```


I then tested for the correlation between Occupational Hazards and the Level of cancer. This time I used Pearson correlation and created another scatterplot with a regression line using a different built-in method in R.

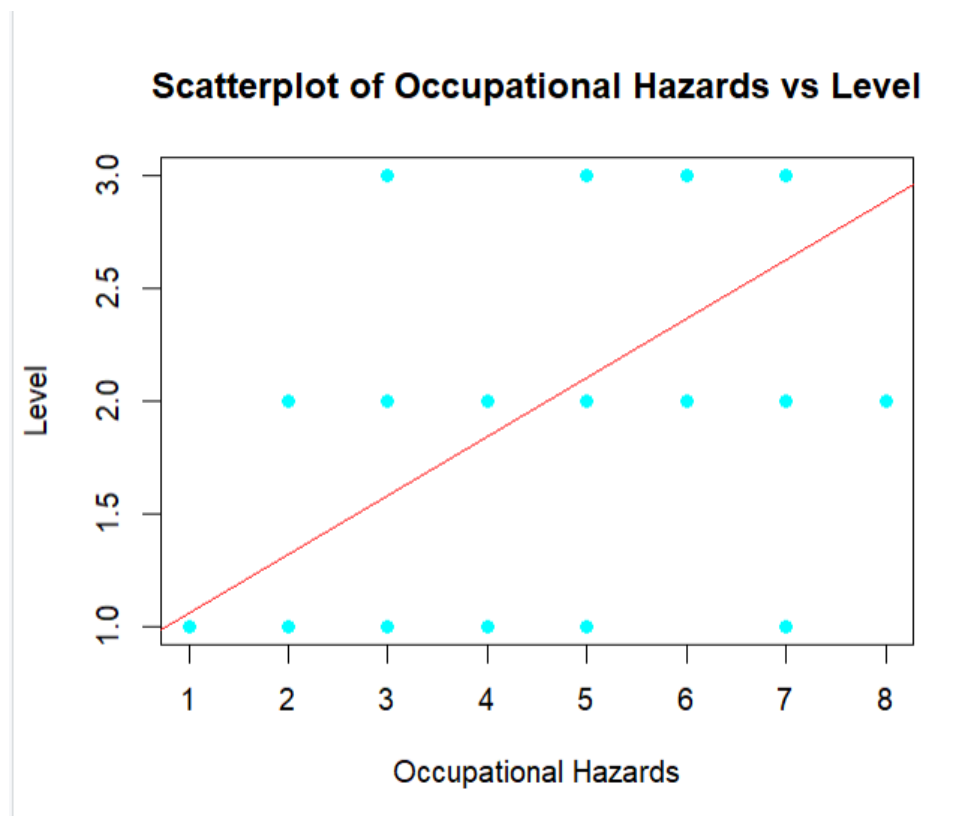
```
> correlation <- cor(df$Occupational.Hazards, df$Level, method = "pearson", use = "complete.obs")
> print(correlation)
[1] 0.6732549
> savehistory("~/testdir/Rhistorystats.Rhistory")
> cor_test <- cor.test(df$Occupational.Hazards, df$Level, method = "pearson")
> print(cor_test)
```

Pearson's product-moment correlation

```
data: df$Occupational.Hazards and df$Level
t = 28.765, df = 998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.6378853 0.7057903
sample estimates:
      cor
0.6732549
```

```
> plot(df$Occupational.Hazards, df$Level,
+       main = "Scatterplot of Occupational Hazards vs Level",
+       xlab = "Occupational Hazards", ylab = "Level",
+       pch = 19, col = "cyan")
> abline(lm(df$Level ~ df$Occupational.Hazards), col = "red")
```

The Pearson correlation score is ~0.67 which indicates a positive relationship between the Occupational hazards and Level of cancer. Since the correlation score is >0.5, there is a direct relationship between Hazards and Level.



Comparing between Alcohol Use and the level

The correlation value = ~0.72 indicates a strong positive correlation between Alcohol.use and Level, so higher alcohol use is directly related to a higher level of cancer. The extremely small p-value further proves this. Additionally, the confidence interval is 95% (95% confidence that the correlation value lies between 0.687 and 0.747), further proving that there is a real, strong, positive correlation.

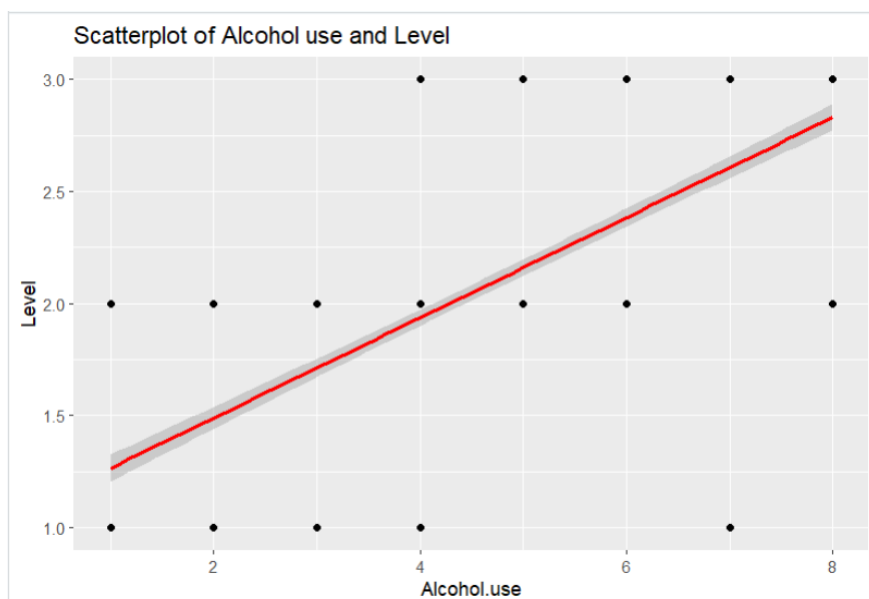
```
> cor(df$Alcohol.use, df$Level, method = "pearson")  
[1] 0.7187103  
> cor.test(df$Alcohol.use, df$Level, method = "pearson")
```

Pearson's product-moment correlation

```
data: df$Alcohol.use and df$Level  
t = 32.654, df = 998, p-value < 2.2e-16  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 0.6873418 0.7474028  
sample estimates:  
      cor  
0.7187103
```

Ggplot

```
> ggplot(df, aes(x = Alcohol.use, y = Level)) +  
+   geom_point() +  
+   geom_smooth(method = "lm", col = "red", se = TRUE) +  
+   labs(title = "Scatterplot of Alcohol use and Level", x = "Alcohol.use", y = "Level")  
`geom_smooth()` using formula = 'y ~ x'
```



Linear Regression Model:

I wanted to compare between all the three variables I selected for my statistical analysis to the target label at the same time in order to identify which had the highest statistical significance and affected the target variable the most. In terms of my project, that would mean identifying which factor between Alcohol.use ,OccuPational.Hazards , or Air.Pollution has the largest impact on the level of cancer an individual has.

```
> model <- lm(Level ~ Alcohol.use + OccuPational.Hazards + Air.Pollution, data = df)
>
>
> summary(model)
```

Call:

```
lm(formula = Level ~ Alcohol.use + OccuPational.Hazards + Air.Pollution,
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.7219	-0.4496	0.1769	0.5392	1.1444

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.77456	0.04870	15.905	< 2e-16	***
Alcohol.use	0.10126	0.01672	6.058	1.96e-09	***
OccuPational.Hazards	0.09102	0.01741	5.229	2.08e-07	***
Air.Pollution	0.10022	0.01298	7.722	2.79e-14	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5472 on 996 degrees of freedom

Multiple R-squared: 0.551, Adjusted R-squared: 0.5497

F-statistic: 407.5 on 3 and 996 DF, p-value: < 2.2e-16

I first identified my target label and variables to fit them into a multiple Linear Regression Model. Then I ran summary() to get some statistical values related to the data. I started by looking at the coefficients. Alcohol.use turned out to be the highest with a value of ~0.101, which also means it has the most effect on the Level. Then there's Air.Pollution which came in as a very close second with a value of ~0.1 and OccuPational.Hazards came last which also means it has the least effect on the Level. The overall p-value is also small, meaning the variables are significant to this analysis.

Validating/ Testing the model:

Since I chose Linear Regression, I decided to validate my model by testing for multicollinearity between my selected variables. I did that by installing the car package, accessing and checking the Variance inflation factor or VIF.

```
> library(car)
Loading required package: carData
```

```
Attaching package: 'car'
```

```
The following object is masked from 'package:dplyr':
```

```
recode
```

```
> vif(model)
      Alcohol.use Occupational.Hazards      Air.Pollution
      6.402626      4.493093      2.317353
```

The results I got for Air.Pollution and Occupational.Hazards are <5 thus showing moderate multicollinearity which is acceptable for the purposes of My analysis. Meanwhile, Alcohol.use had a value of 6.4 which is >5 showing stronger multicollinearity, so I tested for correlation again to see which variable it is highly correlated with.

```
> cor(df[c("Alcohol.use", "Occupational.Hazards", "Air.Pollution")])
      Alcohol.use Occupational.Hazards Air.Pollution
Alcohol.use      1.0000000      0.8787859      0.7472926
Occupational.Hazards 0.8787859      1.0000000      0.6089245
Air.Pollution    0.7472926      0.6089245      1.0000000
```

As shown above, Alcohol.use has significant correlation with Occupational.hazards. Initially, I thought about using Regularization to bring the IVF value down but realized that this correlation makes sense and does not negatively affect my analysis. It provides an interesting insight which is that people exposed to more occupational hazards experience more stress and probably drink more alcohol.

Conclusion:

Based on my findings, I determined that I could accept my Hypotheses and reject the null hypothesis. Even though I was able to prove my hypotheses and to demonstrate a linear relationship between my selected variables, there are not enough details about the data that helps us better understand the nature of these variables. For example, what kind of occupational risks are these individuals exposed to? Is the air pollution exposure related to where they live or work? Similarly, I think the size of the dataset is relatively small therefore, I believe if I had a larger dataset with more patients, I'd be able to demonstrate the correlation better and prove it further. All in all, there is indeed a significant positive relationship between the Level of cancer a patient has and the exposure to Air pollution, occupational risk, and Alcohol use, with Alcohol use having the largest effect. To address these problems and understand their sources better, people first need to become more educated about it. Healthcare organizations and governments can do that by setting up awareness campaigns and by making useful information easily accessible to the public. These findings additionally show that this data can also tremendously help in early cancer detection and classifying high risk groups.

Reference:

McGinnis, A. (2022). *Real-World Data from Taiwan Shows Stage Shift Has Improved Lung Cancer Survival Rates*. [online] ILCN.org (ILCN/WCLC). Available at: <https://www.ilcn.org/real-world-data-from-taiwan-shows-stage-shift-has-improved-lung-cancer-survival-rates/>.